

LENANA SCHOOL 2025 PREDICTION



POSSIBLE KCSE EXAM COMPUTER STUDIES

(THEORY)

451/1

PAPER 1

TIME: 2½ HOURS

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Kenya Certificate of Secondary Education.

INSTRUCTION TO CANDIDATES

- *Write your name and index number in the spaces provided above*
- *This paper consists of **Two** sections **A** and **B***
- *Answer **ALL** questions in section **A***
- *Answer question 16 and any other **THREE** questions from section **B***
- *All answers should be written in the spaces provided on the question paper*

FOR EXAMINER'S USE ONLY

SECTION	QUESTIONS	CANDIDATE'S SCORE
A	1 -15	
B	16	
	17	
	18	
	19	
	20	
	TOTAL SCORE	

SECTION A (40 MKS)

Answer all questions in this section.

1. State the software technological differences between the second generation and the third generation computers [2 marks]

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2. State **two** ways of ensuring proper ventilation in a computer room [2 marks]

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- 3. Convert the decimal number 20.375 to it binary number system equivalent [3 marks]**

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5. State **three** ways of transforming a picture embedded in a Desktop Publishing programme in order to fit in a designated space on a computer. **[3 marks]**

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6. Distinguish between a line printer and a page printer as used in computers. **[2 marks]**

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7. Describe virtual reality as used in computers. **[2 marks]**

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8. State the function of each of the following features of an email software

a) Inbox **[1 mark]**

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b) Draft **[1 marks]**

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9. a) Explain the term toggle key as used in computer keyboard **[2 marks]**

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b) List two examples of toggle keys on a keyboard

[2 marks]

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10. State the type of error that would occur in data processing for each of the following cases:

a) Entering the number 315 instead of 351

[1 marks]

b) Entering the text “Kwys” instead of “Keys”

[1 mark]

11. State four factors to consider when selecting an input device for use in a computer room.

[4 marks]

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12. State three functions of Un-Interruptible Power Supply (UPS)

[3 marks]

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13. Distinguish between a page break and a column break as used in a word processor. [3 marks]

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14. State **three** factors to consider when selecting an operating system to install in a computer.

[3 marks]

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15. Differentiate between analogue signal and digital signal as used in computers. [2 marks]

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SECTION B (60 marks)

Answer ALL questions from this section in the spaces provided.

16. a) Explain the term dry running as used in program development.

[2 marks]

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b) Describe **three** properties of an algorithm. [3 marks]

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- c) Write an algorithm using both pseudocode and flow chart to print the sum and average of numbers from 1 to n where n is provided by user and $n \geq 1$. **[10 marks]**

17. a) State **four** outcomes that may result from using incorrect requirement specifications during system development. [4 marks]

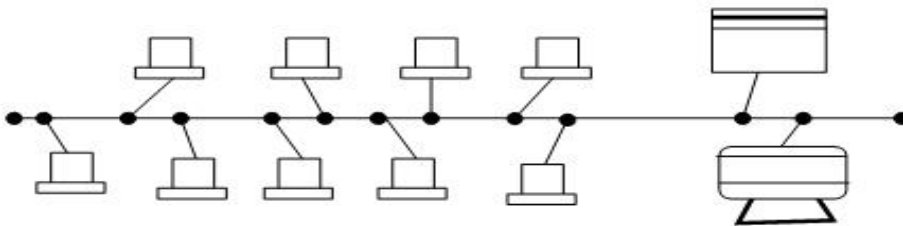
b) A school opted to use direct change over approach when installing a new system. Explain three challenges that the school may face as a result of this approach. [6 marks]

c) State **three** reasons that may lead an organization to install an intranet. [3 marks]

d) The figure below shows a coaxial cable. State the function of the part labelled A. [2 marks]



18. Use the figure below to answer the questions that follow:



a) Give the name of the above network topology [1 mark]

b) State three disadvantages of the above topology [3 marks]

c) A publisher intends to use a desktop publishing programme to create a publication which is to have many graphics. State three ways in which the graphics may be acquired for this purpose. [3 marks]

d) Interpret each of the following spreadsheet formula:

i) =count if (D2:D9,">" B4) [2 marks]

ii) =sumif (B3:B11,"4M", D3:D11) [2 marks]

e) i) The following are the fields of products table in a database created to store records of products manufactured in factory: Product number

- Product name
- Price
- Product description

I. State the appropriate data type for each field [4 marks]

19. a) Explain **three** features of a graphical user interface as used in computers. [6 marks]

b) State the function of each of the following computer keyboard keys:

i) Caps Lock [1 mark]

ii) Shift [1 mark]

iii) Home [1 mark]

c) Using two's complement notation, determine the value of the operation $25_{10} - 29_{10}$. [6 marks]

20. a) Distinguish between a computer drive and computer driver. [2 marks]

b) State **two peripheral** devices connected to the computer via PS/2 [1 mark]

c) Jane encountered desktop computer with the following features

- 3.0 USB ports
- 4.0 GB primary storage
- 19" TFT VDU
- 500 SATA HDD
- Linux 7.1
- AVG antivirus

i) Give 3 reasons for wide use of USB gadgets in the society today. [3 marks]

ii) What is the meaning of the following specifications:

i. 4.0 GB primary storage [1 mark]

ii. 19" TFT VDU [1 mark]

iii. 500 SATA HDD [1 mark]

iv. Linux 7.1 [1 mark]

iii) Give **three factors** to consider when selecting an **OS** to install in your computer. [3 marks]

iv). Describe **two symptoms** of a computer infected by virus. [2 marks]